School of Computer Science and Engineering
(Computer Science & Engineering)
Faculty of Engineering & Technology
Jain Global Campus, Kanakapura Taluk - 562112
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"PROSTATE CANCER PREDICTION USING DEEP LEARNING"

Submitted in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

Submitted by

DNISHITHA - 23BTRCL065, SUMITH REDDY- 23BTRCL171, DABBARA MAHESH - 23BTRCL177, SADWIK - 23BTRCL170, TEJESJWARA REDDY 23BTRCL253

Under the guidance of

Dr. S.kanithan
Assistant professor, Computer Science and Engineering,
School of Computer Science & Engineering, Faculty of Engineering & Technology
JAIN (Deemed to-be University)

Department of Computer Science and Engineering
School of Computer Science & Engineering
Faculty of Engineering & Technology
Jain Global Campus, Kanakapura Taluk - 562112
Ramanagara District, Karnataka, India

CERTIFICATE

This is to certify that the project work titled **"PROSTATE CANCER PREDICTION USING DEEP LEARNING"** is carried out by **Dinishitha (23BTRCL065),Sumith reddy (23BTRCL171), Dabbara Mahesh(23BTRCL177), Sadwik (23BTRCL170), Tejeshwara reddy (23BTRCL253)** bonafide student(s) of Bachelor of Technology at the School of Engineering & Technology, Faculty of Engineering & Technology, JAIN (Deemed-to-be University), Bangalore in partial fulfillment for the award of degree in Bachelor of Technology in Computer Science and Engineering, during the year **2025-2026**.

Dr. S.kanithan Assistant professor Computer Science and Engineering, School of Computer Science & Engineering, Faculty of Engineering & Technology JAIN (Deemed to-be University)	**Dr. V Chandrashekar** Computer Science and Engineering, School of Computer Science & Engineering, Faculty of Engineering & Technology JAIN (Deemed to-be University)	**Dr. Sankar K** PCL Cordinator, Computer Science and Engineering School of Computer Science & Engineering, Faculty of Engineering & Technology JAIN (Deemed to-be University)
Date:	Date:	Date:

Name of the Examiner

Signature of Examiner

1.

2.

DECLARATION

We , **DINISHITHA (23BTRCL065), SUMITH REDDY (23BTRCL170), DABBARA MAHESH (23BTRCL177), SAWDIK (23BTRCL170), TEJESHWARA REDDY(23BTRCL253)** students of VI Semester B.Tech in **Computer Science and Engineering (Artificial Intelligence and Machine Learning)**, at School of Engineering & Technology, Faculty of Engineering & Technology, **JAIN(Deemed to-be University)**, hereby declare that the Project work titled **"Prostate Cancer Prediction Using Deeplearnig "** has been carried out by us and submitted in partial fulfillment for the award of degree in **Bachelor of Technology in Computer Science and Engineering** during the academic year **2025-2026**. Further, the matter presented in the work has not been submitted previously by anybody for the award of any degree or any diploma to any other University, to the best of our knowledge and faith.

Name: Dinishitha USN :23BTRCL065	Signature
Name: Sumith Reddy USN :23BTRCL171	Signature
Name: Dabbara Mahesh USN :23BTRCL177	Signature
Name: Sadwik USN:23BTRCL170	Signature
Name: Tejeshwara Reddy USN:23BTRCL253	Signature

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Signature of Student(s)

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Karan Vasant Nakum	B150438541
Gopal Sanjay Paraskar	B150438545
Pavan Shatrughna Patil	B150438548
Hrushikesh Sadanand Rajankar	B150438553

ABSTRACT

The infectious illness known as Pneumonia is regularly a result of contamination because of a bacterium in the alveoli of the lungs. While an infected tissue of the lungs has an infection, it builds up pus in it. To find out if the patient has those illnesses, professionals perform bodily exams and diagnose their patients through Chest X-ray, ultrasound, or biopsy of lungs. Misdiagnosis, erroneous treatment, and if the disease is overlooked will result in the patient's lack of lifestyle. The progression of Deep learning contributes to aid in the decision-making procedure of specialists to diagnose sufferers with these illnesses. The look employs a bendy and efficient technique of deep learning of applying the model of CNN in predicting and detecting a patient unaffected and affected with the sickness using a chest X-ray photograph. The take a look at utilising an accrued dataset of 20,000 images using a 224x224 photograph decision with 32 batch length is applied to prove the overall performance of the CNN model being educated. The trained-version produced an accuracy charge of 95% at some point of the overall performance training. Based on the end result of the experiment carried out, the research study can detect and are expecting COVID-19, bacterial, and viral-pneumonia sicknesses based totally on chest X-ray photos.

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CHAPTER 1

INTRODUCTION TO PNEUMONIA DETECTION USING CNN AND VGG16

Pneumonia impacts all the elderly and young people everywhere. With the high growth in the popularity of neural networks, engineers and researchers have been able to find state-of-the-art products for computer vision. Artificial Intelligence helps us to automate analysis techniques, which is only possible now because of the technology of Deep Learning. Exposure to Pneumonia is quite high for many people, mainly in economically underdeveloped and developing countries where the majority are deprived of a nutritious diet. The World Health Organisation states that more than 4 million untimely deaths per year occur from diseases caused by air pollution.

The purpose of this project is to build an AI network, which takes the pixel values as input for a given X-Ray image and then proceeds to perform linear operations and activations on each of them. Then by taking all the above operations, and then multiplying them with each layer within the neural network, and the number of nodes. Suddenly you have millions of operations. By applying determination, they can be more efficient in these tasks. The scope is to develop a model that will identify whether a patient is having Pneumonia or not bypassing the chest X-Ray images through the Deep Learning model

1.2 MOTIVATION BEHIND PROJECT TOPIC

The main motivation behind this research is to identify Pneumonia just by using the X-Ray images of the patients. as doctors must do a lot of certain tests to identify if the patient has Pneumonia or not. To solve the cumbersome problem, an ensemble of two deep learning models is developed, to make the work of the doctors simpler.

1.3 AIM AND OBJECTIVE OF THE WORK

Aim:

To provide an efficient and effective solution over the conventional way of detecting pneumonia disease using VGG16 of CNN

Objectives:

- To learn different biomedical terms related to pneumonia disease.
- To learn the different scenarios of pneumonia disease (viral or bacterial).

- To learn different methods of data acquisition. to know about different image processing pre-trained models.
- To build a web application to analyse Pneumonia disease using chest X-ray.

CHAPTER 2

Literature Survey

2.1 Vandecia Fernandes et al. **[1]** Proposed**,** developing a reliable version to detect COVID-19 from X-ray images offers a number of challenges. First of all, there are a limited number of images to be had. The pandemic has simplestly been spreading for a few months and now not many datasets have been gathered and shared for the benefit of researchers. Secondly, there may be an urgent need for scientific researchers to develop a wise device for the fast detection and deep expertise of infectious diseases from chest X-ray pictures, which could illustrate the damage to the human body. After investigating the presently available technology and the demanding situations we are going through, it's concluded that a transfer learning of technique is possible and realistic for this study's problem. Our research technique is to train deep learning models at the available images of pneumonia, and contamination in a single or both lungs which may be resulting from microorganisms, viruses, or fungi. The infection causes inflammation inside the air sacs in the lungs, which might be known as alveoli. The alveoli fill with fluid or pus, making it hard to respire the system takes benefit of transfer learning on properly-studied of deep learning to know the type of virus and validates the models on a big quantity of data sets, and finally transfers the models and insights learned in training and validation to a brand new data set in a comparable area, which, in this have a look at, is a new infectious fast-growing sickness, COVID-19 or coronavirus. The proposed technique solves the tough problem in deep learning, this is, how to construct a dependable model primarily based on a scarce information set, which is not well studied and there are unknown features inside the statistics set.

2.2 Hongen Lu et al.**[2]**Deep CNNs indeed acquire higher performance with the use of a huge dataset as compared to a smaller one. Granting there is an enormous range of infected COVID-19 patients globally, however, the variety of publicly available chest X-ray photos online is insignificant and dispersed. For this reason, the authors of this work have defined a reasonably huge dataset of COVID-19 infected chest X-ray images even though normal and pneumonia photographs are right away on hand publicly and applied in this study.

2.3 Sammy V. Militante et al.**[3]** Proposed**,**The observations employ flexible and efficient approaches of deep learning by making use of six models of CNN in predicting and spotting whether the patient is unaffected or affected by the disease employing a chest X-ray picture. GoogLeNet, LeNet, VGG-16, AlexNet, StridedNet, and ResNet-50 models with a dataset of 28,000 images and using a 224x224 decision with 32 and sixty-four batch sizes are implemented to verify the performance of every version being educated. The study likewise implements Adam as an optimizer that continues an adjusted 1e-four learning rate and an epoch of 500 hired to all the models. Both GoogLeNet and LeNet acquired a ninety-eight% price, VGGNet-sixteen earned an accuracy charge of ninety-seven%, AlexNet and StridedNet model acquired a ninety-six% whilst the ResNet-50 version acquired 80% throughout the training of models. GoogleNet and LeNet fashions performed the best accuracy rate for overall performance training. The six models identified had been capable of hitting upon and are expecting pneumonia sickness including a wholesome chest X-ray.

2.4 Nanette V. Dionisio et al.**[4]**Proposed, classify the three varieties of X-rays, image writer used the ensemble method at some stage in prediction, each picture is surpassed through the type layer where they checked whether an image is COVID-19, pneumonia, or ordinary.

Sr. no	Paper Allocation	Author	Year	Methods
1	Bayesian convolutional neural network estimation for paediatric pneumonia detection and diagnosis	Vandecia Fernandes et al.	2021	Bayesian convolutional neural network
2	Transfer Learning from Pneumonia to COVID-19	Hongen Lu et al.	2020	Transfer Learning
3	Pneumonia and COVID-19 Detection using Convolutional Neural Networks	Sammy V. Militante et al.	2021	Convolutional Neural Networks
4	Pneumonia Detection through Adaptive Deep Learning Models of Convolutional Neural Networks	Nanette V. Dionisio et al.	2021	Deep Learning Models

Table 1